Questions: Planning the Creative Process

Analytical Questions

1. Explain the exploration-exploitation trade-off in creative planning using the Active Inference concepts of epistemic and pragmatic expected free energy. Why does the optimal balance shift over the course of an inventive project?

2. How does the "double diamond" design model map to the Active Inference framework? Describe what happens to the agent's policy distribution (entropy) during the divergent and convergent phases.

3. Define the difference between parametric uncertainty and structural uncertainty using Active Inference terms. Why does creative work involve more structural uncertainty than conventional engineering, and what planning implications does this have?

4. Analyze stage-gate processes as formalized expected free energy evaluations. At each gate, what information is being compared to what criteria, and how does this prevent both premature convergence and indefinite exploration?

5. Why does planning depth (temporal horizon) need to match prediction reliability? What goes wrong when an inventor creates a detailed 12-month plan for a project where predictions are only reliable for the next two weeks?

6. Explain the concept of "information-maximizing milestones." Why should the highest-impact, most-testable uncertainties be addressed earliest in the plan, and how does this relate to expected free energy minimization?

7. Herbert Simon's concept of "satisficing" suggests that selecting a good-enough solution and iterating may outperform searching for the optimal solution. Explain this using Active Inference — under what conditions does the cost of continued exploration exceed the expected benefit?

8. SpaceX used a "fail-fast" planning structure for rocket landing, attempting landings as secondary objectives on primary missions. Analyze this as an Active Inference planning

strategy. How does the fail-fast structure manage the trade-off between information gain and catastrophic loss?

9. Compare adaptive planning (revising the plan as evidence arrives) with fixed planning (maintaining the original plan regardless of evidence). Use the Active Inference concept of model updating to explain why adaptive planning is superior for creative projects but may be inferior for routine projects.

10. The Dyson bladeless fan case study shows the team extending their timeline by six months when testing revealed acoustic problems. Analyze this as adaptive replanning. Was this a planning failure (they should have predicted the problem) or a planning success (they responded appropriately to structural uncertainty)?

Applied Questions

1. Assess whether your current invention project should be in an exploration or exploitation phase. List the evidence supporting your assessment and describe what you would need to see to switch to the opposite phase.

2. Classify the top five uncertainties in your invention project as parametric (known model, unknown values) or structural (unknown model structure). For each, describe how you would reduce it.

3. Design a three-stage plan for your invention project with explicit gate criteria. At each gate, specify what evidence would lead to continue, pivot, or stop decisions.

4. Reorder your project milestones to maximize information gain. Place the highest-impact, most-testable uncertainties first. How does this change the sequence compared to your intuitive plan?

5. Create a multi-horizon plan: detailed actions for the next two weeks, objectives for the next two months, and direction for the next six months. How does the level of detail change across horizons?

6. Identify three failure modes for your project plan. For each, describe an early warning signal and your planned response. Would any of these failure modes require a pivot rather than a fix?

7. Apply the convergence criteria from this module to your current situation. Are you seeing solution clustering, diminishing returns, or constraint satisfaction? What does this tell you about whether to continue exploring or commit?

8. Describe a time in your inventive work when you converged too early or explored too long. With hindsight, what signal did you miss that would have told you to switch phases earlier?

9. Design a "plan review" practice for your invention project — a regular interval at which you will compare the plan's predictions against actual outcomes and revise accordingly. How often should you review? What metrics will you track?

10. Write a one-page "Inventive Project Plan" for your current project that includes: current stage assessment, key uncertainties ordered by impact and testability, stage-gate plan with decision criteria, multi-horizon detail, and planned review dates. This should be a living document you can actually use.